

Sravya Aravapalli

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EDUCATION

Carnegie Mellon University

May 2028

Bachelor of Science in Computer Science, Minor in Machine Learning

Pittsburgh, PA

Courses: Computer Systems, Machine Learning, Data Structures & Algorithms, Functional Programming, Discrete Mathematics, Probability, Linear Algebra; **Current:** Distributed Systems, Generative AI

Organizations: Carnegie Mellon Racing, Society of Women Engineers, Alpha Chi Omega: Kappa Nu

TECHNICAL SKILLS

Languages: Python, Java, C++, C, TypeScript, JavaScript, SML, HTML/CSS

Technologies: Git, NumPy, PyTorch, Pandas, Sklearn, React, Next.js, PostgreSQL, Prisma, Tailwind CSS

EXPERIENCE

Axon

May 2026 – Aug. 2026

Firmware Engineering Intern

Boston, MA

- Developed C/C++ firmware for an embedded device to offload IMU data over LTE, implementing data serialization, buffering, and MQTT-based transmission for prototype data collection
- Designed concurrent data collection and transmission workflows using RTOS threading while working within embedded device constraints; flashed and tested firmware directly on hardware
- Built Python analysis scripts to identify patterns and detect events across around 10 hours of collected IMU data, achieving 90% event-detection accuracy against human-labeled ground truth

Carnegie Mellon University

Aug. 2026 – Present

Teaching Assistant

Pittsburgh, PA

- Leading recitations and office hours, assisting in developing and grading assignments and exams, and supporting review sessions for 15-151 Mathematical Foundations for Computer Science

CMU Computer Science Department

Jan. 2026 – May 2026

Undergraduate Researcher

Pittsburgh, PA

- Worked under Professor Reid Simmons on interactive robotics system design and development.
- Led the migration of Gamebot's C++ Interaction Manager to Python for an autonomous Scrabble-playing robot used in human-computer interaction research, redesigned its rule-based priority action pipeline to improve scalability and maintainability.

University of Minnesota

June 2025 – Aug. 2025

Undergraduate Researcher

Minneapolis, MN

- Selected into 'AI for Earth Summer School', focused on applying AI and machine learning to solve challenges in domains such as climate change, agriculture, and sustainability
- Developed an LSTM based deep learning model to predict NEE, GPP, and RECO Net Ecosystem Exchange (NEE), Gross Primary Productivity (GPP), and Ecosystem Respiration (Reco) metrics, in order to assess carbon dioxide fluxes and forecast ecosystem sustainability at several global sites
- Achieved an RMSE of 0.53 for GPP prediction, corresponding to around 5% of its daily flux range

SBPL Research Lab, CMU Robotics Institute

Feb. 2025 – Apr. 2025

Undergraduate Researcher

Pittsburgh, PA

- Built and optimized a 2D occupancy grid-based collision checker using rectangular footprints for quadrupedal motion planning research under Rishi Veerapaneni and Professor Maxim Likhachev.
- Implemented batched geometric transformations and collision validation using NumPy vectorization (einsum, broadcasting) and linear interpolation for efficient path checks.

PROJECTS

Astro Sandbox | TypeScript, Next.js, React, Three.js, PostgreSQL

Aug. 2025 – Nov. 2025

- Developed a real-time orbital mechanics simulator with interactive 3D Three.js visualization, Web Worker-based physics computation, and Prisma ORM/PostgreSQL backend APIs